Earth Embeddings for Political Science: How They Work, What They Measure, What Comes Next

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Abstract

Earth embeddings are compact vector representations of geographic locations, produced by deep neural networks trained on satellite imagery and other geospatial data. A single vector per location encodes its built environment, infrastructure, land use, and spatial organization, and can be queried by coordinates and merged with any georeferenced dataset. The technology is very new and has so far been tested almost entirely by computer scientists. This paper reviews it in three parts: how earth embeddings are produced, where they fit in the longer tradition of satellite-based measurement in social science, and one conceptual direction they open for empirical work. On the applied side, the paper argues that physically grounded outcomes, with public goods provision and physical infrastructure as the most natural examples, are where the technology has the most potential in political science today. On the conceptual side, it sketches how embeddings could serve as a new ingredient for spatial estimation, alongside the distance-based tools that spatial econometrics and political methodology have wrestled with for decades.

1 Introduction

Political scientists care about geography. The discipline studies how individuals behave under the authority of a state and its institutions, how states compare to other states, how policies diffuse across jurisdictions, and how local conditions shape political behavior. Yet the discipline's empirical toolkit treats geography almost entirely as a structural nuisance. Countries are rows in a panel. Municipalities are polygon boundaries. Neighborhoods, when they appear at all, enter the analysis as fixed effects or dummy variables. The rich, continuous reality of place, its built environment, its infrastructure, the physical organization of its communities, tends to be reduced to an identifier or a polygon.

Earth embeddings are a technology that makes it possible to stop reducing geography in this way. They are vector representations of geographic locations produced by AI models trained on satellite imagery and related geospatial data. A compact vector for a geographic unit encodes information about that unit's built environment, road network, vegetation, land use, and spatial organization, information that would otherwise require specialized remote sensing pipelines to extract. Three properties make these vectors useful for empirical research: *generality*, the same vector can serve as input to models predicting poverty, crime, or infrastructure quality; *accessibility*, pre-computed embeddings can be queried by coordinates and merged with any georeferenced dataset; and *richness*, they encode far more about a location than any single hand-crafted proxy such as nighttime lights.

A brief note on the nature of this paper. No single discipline owns the set of ideas discussed here. Earth embeddings come from computer vision and self-supervised learning. Their use as proxies for socioeconomic outcomes draws on development economics, remote sensing, and, in a handful of applied papers, political science. The conceptual move in the last part of the paper, about how researchers represent the way geographic units relate to one another in empirical models, belongs to spatial econometrics and, increasingly, to political methodology, where it has long been a topic of active debate. Bringing these literatures together is inherently a cross-disciplinary exercise, and the paper reflects that. The contribution is to pull from three literatures that have developed largely in isolation and to read them as one body of work.

The paper is organized around three arguments, in order.

Section 2 explains how earth embeddings are produced. It covers the embedding concept, the production pipeline from satellite imagery through Vision Transformers and self-supervised training, the four functions that Klemmer et al. (2025) use to organize what these vectors can do, the model ecosystem that currently produces them, and what an embedding actually looks like to an applied researcher. The section is deliberately technical. A reader who knows nothing about computer vision should come out of it able to describe what a 64-dimensional AlphaEarth vector *is*, where it comes from, and why its geometry encodes anything useful.

Section 3 turns to applied use. It places earth embeddings inside the longer trajectory of satellite-based measurement in social science and acknowledges how recently the technology emerged. The section suggests that, given how earth embeddings are constructed, physically grounded outcomes, with public goods provision and physical infrastructure as the most natural examples, are where the technology has the most potential in political science today, and it reviews the empirical evidence available so far.

Section 4 is the conceptual jump. Social science does not, in the end, ask prediction questions; it asks how X causes Y for specific units i, and whether that causal effect might be heterogeneous in the geographic character of each unit. The section introduces that question, shows how spatial econometrics and political methodology have historically answered it through the matrix W, reviews what has been learned and what remains unsettled, and discusses how earth embeddings offer a genuinely new ingredient to bring to the problem.

Section 5 discusses limitations of the technology and Section 6 closes the paper.

2 How Earth Embeddings Work

This section explains the technology. It starts from the embedding concept itself, walks through the production pipeline from satellite inputs to a d-dimensional vector, presents the four functions earth embeddings are designed to perform, and closes with the current model ecosystem and what an embedding actually looks like in practice. Klemmer et al. (2025) formalize earth embeddings as a coherent object of research and provide the framework the section draws on throughout.

2.1 The Embedding Concept: From Words to Places

The idea behind an embedding is to represent a complex object as a point in a vector space, arranged so that the geometric relationships between points (distances, directions, clusters) reflect meaningful relationships between the objects they represent.

The concept was popularized in natural language processing. Mikolov et al. (2013) showed that training a neural network to predict words from their contexts produces vector representations whose geometry encodes semantics. The vectors for *king* and *queen* end up near each other. The vector difference between *king* and *man* is similar to that between *queen* and *woman*. Words that appear in similar contexts end up near each other in vector space, even when they never co-occur directly in the training data. The structure that the model has internalized about language emerges as geometric structure in the embedding space.

Earth embeddings apply this principle to geography. An embedding function E maps a location, and optionally a time, into a d-dimensional vector:

$$E : (\text{location}, \text{time}) \to \mathbf{e} \in \mathbb{R}^d.$$

The key property is that the geometry of this vector space is meant to reflect *geographic* structure. Two urban cores with comparable density and road layout end up close together in embedding space, even across continents. A coastal strip and an inland farming plain end up far apart, because their visible characteristics differ substantially. Seasonal variation appears too: summer and winter embeddings for equatorial locations may be nearly identical, while embeddings for highly seasonal regions differ visibly across the year. The embedding, informally, is a learned summary of what a location looks like from above, compressed into a manageable number of dimensions.

2.2 The Production Pipeline

How an earth embedding is produced determines what information it actually contains. The pipeline has three components: input data, model architecture, and training objective. The choices made at each step shape what shows up in the final vector. Figure 1 summarizes the flow.

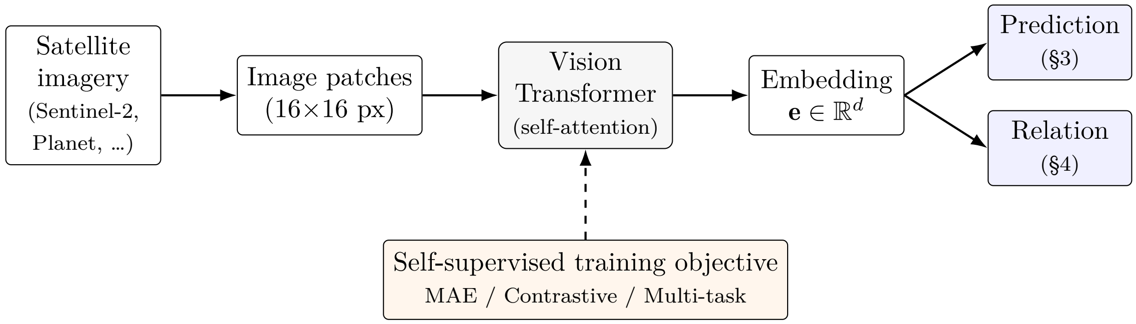

Figure 1: The earth embedding production pipeline. Satellite imagery is divided into fixed-size patches, processed through a Vision Transformer, and reduced to a d-dimensional vector that downstream applications can consume. The training objective (dashed arrow) is the design choice that determines what the embedding actually encodes.

2.2.1 Input Data

Earth embedding models are trained on geospatial observations indexed by location and time. The most common inputs are multi-spectral satellite images from missions like Sentinel-2 (10-meter spatial resolution, five-day revisit cycle, freely available) or commercial providers like Planet (3 to 5 meter resolution, daily coverage).[1] Some models also incorporate radar data (Sentinel-1, robust to cloud cover), geotagged text, point-of-interest databases, or demographic variables. The spatial resolution and revisit frequency of the input data determine the finest scale at which embeddings can distinguish between locations; the spectral richness of the input determines what kinds of physical features can in principle be encoded. A model trained on RGB imagery cannot represent vegetation moisture; one trained with a near-infrared channel can.

2.2.2 The Vision Transformer Architecture

The dominant architecture for processing satellite imagery is the *Vision Transformer*, or ViT (Dosovitskiy et al. 2021). A ViT divides an input image into fixed-size patches, typically 16 by 16 pixels. Each patch is projected into a vector through a linear transformation, called a *patch embedding*. Positional information is added to each patch embedding so that the model knows where each patch sits within the image. The resulting sequence of patch embeddings is then processed through a stack of *transformer layers*, the same architectural family that underlies large language models. Each layer applies a self-attention mechanism that lets every patch "look at" every other patch, allowing the model to integrate context from across the whole image. The output of the stack consists of contextualized patch embeddings along with a global token (often called the CLS token) that summarizes the entire image as a single vector. This global token, or some pooled

[1] A quick clarification for readers outside remote sensing. A satellite image is not an image in the everyday sense of a JPEG on a phone. It is a *raster*: a grid of cells, essentially a matrix, where each cell records one or more numerical measurements of light reflected from a patch of ground. A photograph has three channels (red, green, blue). A Sentinel-2 scene has thirteen channels, covering not only visible light but also near-infrared, short-wave infrared, and narrower bands used for vegetation, water, and cloud detection. "Multi-spectral" just means the raster has more than the usual three channels. The number and wavelength of those channels is what the model ultimately gets to see, which is why choosing the input data is already a substantive modeling decision.

function of all the patch tokens, is what serves as "the embedding" for the image. Convolutional neural networks (ResNet, EfficientNet) remain a viable alternative to ViTs, particularly for tasks that benefit from explicit spatial hierarchies, but the transformer has become the default in recent geospatial foundation models.

2.2.3 Training Objectives: What the Embedding Learns

The training objective is the most consequential design choice in the pipeline. It determines what the model has to internalize in order to do well at its training task, and therefore what shows up in the embedding. Three approaches dominate. Table 1 compares them.

Table 1: Three self-supervised training objectives used to produce earth embeddings. The objective determines what the embedding learns to represent.

Objective	**Mechanism**	**What the embedding learns**	**Example models**
Masked autoencoding (MAE)	Hide a large fraction of patches; reconstruct them. Loss: $\|x - \hat{x}\|^2$.	Visual structure and spatial regularities of imagery.	Clay, Prithvi
Contrastive learning	Pull together embeddings of co-located observations; push apart embeddings of distant ones.	Location identity from recurring observable patterns.	SatCLIP, GeoCLIP
Supervised multi-task	Predict several known location-specific outcomes simultaneously from a shared backbone.	Whatever features are jointly predictive of the supervised targets.	AlphaEarth Foundations

The takeaway from this comparison is that the training objective is, in practice, a choice about what the embedding "knows." A model trained with masked autoencoding learns to represent the visual texture of landscapes, because it has to fill in missing tiles. A model trained with contrastive learning learns to represent location identity, because that is what its loss rewards. A model trained with supervised multi-task objectives learns to represent whatever happens to be predictive of the supervised targets at once. Two embeddings trained on identical satellite imagery, but with different objectives, will have different geometric structure and different downstream uses.

The supervised multi-task case is the least self-explanatory of the three, so a concrete example helps. AlphaEarth Foundations (Brown et al. 2025) is the clearest one, and the model this paper returns to most often in its applied sections. It takes optical imagery from Sentinel-2, radar from Sentinel-1, and a set of auxiliary geospatial layers as input, and produces a single 64-dimensional vector per location from which several structural quantities, among them land cover, temperature, and elevation, can all be recovered with reasonable accuracy. The training loss is a combination of the prediction errors across these tasks. Because the backbone has to serve all of them at once, the resulting vector ends up carrying whatever joint signal is needed for the whole bundle, rather than being tuned to any single outcome. That is what "supervised multi-task" means in practice:

the compression is not arbitrary; it is shaped by the set of targets the model was asked to hit.

2.3 What Earth Embeddings Do: Four Functions

Klemmer et al. (2025) organize the functions of earth embeddings around four properties. The taxonomy is useful both as a vocabulary for what these vectors can do and as a way of seeing why they are interesting beyond computer vision.

Compression distills high-dimensional, multi-modal geospatial data into compact vectors. A satellite image patch containing millions of pixel values across multiple spectral bands becomes a vector of 64 or 256 numbers. What would previously have demanded a remote-sensing pipeline, band-by-band feature extraction, and a specialized workflow per task, now fits inside a lookup table.

Fusion combines different data modalities (optical imagery, radar, text, demographic data) into a single representation indexed by location. This matters when the signal of interest is not reducible to any one modality: a neighborhood is not just its land cover, or just its nighttime lights, or just its text-tagged businesses. It is a combination, and fusion is what allows an embedding to encode the combination rather than forcing the researcher to stack separate pipelines.

Interpolation, available through models that use implicit neural representations, provides embeddings for arbitrary coordinates in continuous space, including locations absent from the training data. This is what allows, for instance, a location encoder to return an embedding for any (lat, lon) pair a researcher cares to query, not just for grid cells that happened to be sampled.

Integration (what Klemmer et al. call *interoperability*) allows embeddings to serve as location tokens that plug into other AI systems. Just as word embeddings became the standard input representation for language models, earth embeddings are beginning to serve as the standard geographic representation for models that need to reason about place.

For political science, the immediately relevant properties are compression and fusion. Compression means the researcher does not need to process raw satellite imagery. Fusion means that heterogeneous data sources are integrated without requiring separate pipelines for each. The other two properties matter for downstream methodological work: interpolation gives a principled way to assign embeddings to arbitrary coordinates, and integration is what allows embeddings to enter, for example, language-model workflows that reason about geographic context.[2]

[2] One caveat about access that is worth naming early. The models that produce these vectors can only be trained by organizations with access to large amounts of satellite imagery and substantial compute, which today means a small number of technology companies and well-resourced research labs. In that sense earth embeddings resemble the large-language-model revolution: ordinary researchers will rarely, if ever, train one from scratch. Fortunately, and unlike the closed frontier of the largest language models, much of the earth-embedding ecosystem is either open-source (Clay) or distributes pre-computed vectors through public APIs (AlphaEarth via Google Earth Engine). The product the applied researcher actually uses is a number, a vector, queried by coordinate, and that number can be used downstream without any of the upstream infrastructure.

2.4 The Model Ecosystem

Several embedding models are now available, and they vary in how they produce embeddings and how they deliver them to users. Table 2 summarizes the main options.

Table 2: The current earth embedding model ecosystem. Two families: explicit feature extractors that process imagery into vectors, and implicit location encoders that return vectors directly from coordinates.

Model	Family	Input	Training	Dim.	Access
AlphaEarth	Explicit	Sentinel-1/2 + auxiliary	Multi-task	64	Google Earth Engine
Clay	Explicit	Multi-sensor imagery	MAE	varies	Open weights
SatCLIP	Implicit (location encoder)	Coords ↔ Sentinel-2	Contrastive	256	Open weights
GeoCLIP	Implicit (location encoder)	Coords ↔ ground-level	Contrastive	varies	Open weights

Two families of models currently dominate the ecosystem. *Explicit feature extractors* process raw geospatial inputs through a neural network and return an embedding for each input image or patch. *Implicit neural representation models*, also called *location encoders*, take a different route: they store geospatial information inside the weights of the model and return an embedding for any (latitude, longitude) query, without processing imagery at query time. The four models in Table 2 span both families. It is worth saying a few words about what distinguishes each.

AlphaEarth Foundations (Brown et al. 2025) is a Google DeepMind model whose defining commitment is extreme compression. It fuses optical imagery (Sentinel-2), radar (Sentinel-1), and auxiliary layers into a single 64-dimensional vector per location, trained with supervised multi-task objectives. What makes it special for applied work, however, is not the model itself but the product: the *Satellite Embedding Dataset*, pre-computed globally at 10-meter resolution for 2017–2024 and served directly through Google Earth Engine. The applied researcher does not run a model; they query a catalog. This is the main reason AlphaEarth is the most accessible option for political scientists today.

Clay (Clay Foundation) is the most flexible model in the set. Trained with masked autoencoding on multi-sensor imagery, it is designed to be agnostic about input resolution and sensor type, from drone photos at 30-centimeter resolution to MODIS at 500 meters. Recent versions add *Dynamic Positional Encoding*, which injects timestamp and coordinates directly into the embedding, and *Matryoshka representations*, which let the researcher use shorter truncations of the vector without losing fidelity. Clay is the right choice when the research design needs to stitch together heterogeneous imagery of the same place across scales or time.

SatCLIP (Klemmer et al. 2025b), developed at Microsoft Research, is the semantic location encoder in the group. It is trained contrastively to align satellite imagery with coordinates, producing a 512-dimensional vector that answers the question "what is this place like?" rather than "where is this?". Two industrial zones on different continents end up close in SatCLIP's embedding space not because they are geographically close but because they share the same kind of place. This makes SatCLIP well-suited for settings where conceptual similarity between locations is itself the quantity of interest.

GeoCLIP (Vivanco Cepeda, Nayak, and Shah 2023) is the oldest of the four and a direct precursor of SatCLIP. It aligns ground-level photographs with coordinates rather than satellite imagery, and so captures a different slice of the world: street-level visual patterns, building facades, signage, what a place looks like at human eye level. GeoCLIP therefore complements satellite-based encoders when the outcome of interest is plausibly shaped by experience at ground level rather than by what is visible from orbit.[3]

2.5 What an Embedding Looks Like in Practice

The easiest way to build intuition for what an embedding *is* is to imagine putting a pin anywhere on Earth and asking the model, "what does this place look like, as a number?" If the reader drops a pin over their own head right now and queries AlphaEarth, they get back a vector: a short list of floating-point numbers that summarizes their immediate surroundings as seen from space. If the reader thinks of the street they grew up on and drops a pin there, they get a different vector. If the reader imagines the paradisiac beach from their dream vacation and drops a pin over it, they get another vector again. The three vectors can be compared with each other, merged with a census table, or fed into a regression. That is the object.

For a political scientist, the most accessible entry point into this ecosystem is a pre-computed embedding database, and AlphaEarth is currently the most usable such database. It provides exactly the kind of interface described above, through Google Earth Engine: the researcher queries embeddings by coordinates, merges them with an analytical dataset, and proceeds with standard statistical methods. The resulting workflow involves no image processing, no deep learning infrastructure, and no remote sensing expertise. What would have required months of specialized pipeline development a decade ago now takes an afternoon of data merging.

A few practical details are worth flagging. The specific dimensionality of the vector depends on the model. AlphaEarth returns 64 numbers per location, which is part of why it is popular in applied work; SatCLIP returns 256; Clay varies; GeoCLIP depends on the backbone. But the shape of the workflow is the same across all of them. The vector is just another covariate block. It can be stacked with census variables, administrative records, or any other tabular data, and fed into

[3] MOSAIKS (Rolf et al. 2021), for reference, is not in the table. It uses a fixed set of random (rather than learned) convolutional features of satellite imagery and combines them with ridge regression. Klemmer et al. (2025) explicitly distinguish it from earth embeddings proper on this basis: the geometry of MOSAIKS features is not shaped by a training objective. Conceptually, MOSAIKS is still the immediate precursor of the foundation-model paradigm and deserves credit for establishing that a single feature set can serve many downstream tasks.

whatever statistical model is appropriate for the substantive question. The technology is accessible. What to *do* with it is the question Sections 3 and 4 take up.

3 Earth Embeddings as a Prediction Tool

This section turns from technology to use. Before reviewing what earth embeddings have been used to measure, it is worth being honest about how early in the technology's life this review is being written, because that framing matters for everything that follows.

3.1 A Note on How New All This Is

Earth embeddings are, at the time of this paper, a very young technology. The framework paper that formalizes them as a coherent object of research, Klemmer et al. (2025), and the flagship queryable database, AlphaEarth Foundations (Brown et al. 2025), are both from 2025. Earlier pieces such as SatCLIP and GeoCLIP go back a bit further, and the intellectual lineage reaches back to earlier satellite-features work such as MOSAIKS (Rolf et al. 2021). But the idea of a general-purpose, pre-trained, queryable geographic embedding as a recognized category is very recent.

Two consequences follow. First, the adoption curve in empirical social science is almost non-existent. There is not, as of this writing, a peer-reviewed political science article that uses earth embeddings as the term is defined here. In empirical economics the picture is the same for earth embeddings specifically, though the older tradition of geospatial proxies, above all nighttime lights, has been in use for well over a decade and is substantial. The gap, in other words, is not that nobody has used satellite-derived variables before, it is that earth embeddings themselves have not yet been used in empirical social science, and it is only a matter of time before someone fills the gap. Second, essentially all of the applied evidence we have on earth embeddings comes from computer science, and there are only a handful of papers: the two main systematic benchmarks discussed below, and a few applied pieces. More is clearly on the way.[4] The technology was built by computer scientists to solve computer science problems, and it has been tested against the kinds of evaluations computer scientists naturally reach for: held-out prediction accuracy on benchmark tasks.

Given how new the technology is, the best way to get a sense of its potential is to look at what has actually been done with it so far. The rest of this section places earth embeddings inside the longer tradition of satellite-based measurement in social science, walks through the applied workflow, and reviews the handful of empirical pieces available to date. Where those pieces point toward a natural home for the tool in political science, the section flags it; where they leave genuine

[4] A practical consequence is that the way earth embeddings are currently evaluated, through benchmark R^2, MSE, and performance across urban indicators, is not the way a political scientist would most naturally evaluate them. Whether a 64-dimensional vector predicts cycling rates is interesting in a computer-science sense; whether it lets us do something *methodologically* new for causal inference is a different question, and one the current benchmarks do not touch.

empirical questions open, it tries to say so plainly.

3.2 Three Generations of Satellite-Based Measurement

The use of satellite-derived data as proxies for socioeconomic outcomes has been one of the most promising methodological directions in quantitative social science of the past two decades, and also one that, despite real progress, has never quite become mainstream. The basic insight is that the physical appearance of a location, its buildings, roads, agricultural patterns, nighttime luminosity, correlates with the conditions of its inhabitants. Where conventional data are absent, infrequent, or unreliable, satellite imagery offers a scalable alternative. The reason the approach has not fully taken off is not that it does not work; it is that, at least in its earlier generations, each application required a heavy, specialized pipeline that few substantive researchers were in a position to build. Three successive generations of methods have implemented this insight, each addressing the limitations of the last. Table 3 summarizes the trajectory.

Table 3: Three generations of satellite-based measurement in social science. Each generation reduces the per-application cost of moving from imagery to a usable feature for empirical work.

Generation	Approach	Canonical example	Limitation
1. Hand-crafted proxies	A single, interpretable index (e.g., nighttime lights) tied to one outcome.	Henderson, Storeygard, and Weil (2012) on GDP from nighttime lights.	A new outcome requires a new proxy.
2. Task-specific deep learning	A convolutional network trained per task; intermediate features are used as covariates.	Jean et al. (2016) on poverty in Sub-Saharan Africa; MOSAIKS (Rolf et al. 2021).	Pipeline must be rebuilt per outcome and per region.
3. General-purpose embeddings	Pre-computed vectors usable across many outcomes and tasks without retraining.	AlphaEarth, SatCLIP, Clay (2023 to 2025).	Construct validity, interpretability, spatial bias remain open.

Nighttime lights have been the most widely adopted first-generation proxy. Henderson, Storeygard, and Weil (2012) showed that satellite-measured luminosity strongly tracks GDP at national and subnational scales, and proposed using lights data to improve growth estimates in countries with poor national accounts. Donaldson and Storeygard (2016) provided an influential overview of how remotely sensed data can serve as outcomes, controls, or instruments in causal designs. The second generation moved from hand-crafted indices to learned ones. Jean et al. (2016) trained a convolutional network to predict nighttime light intensity from daytime satellite imagery and repurposed its intermediate representations to predict consumption in five African countries. The methodological contribution was that deep learning could extract economically relevant features from imagery without the researcher having to specify which visual patterns matter. Rolf et al. (2021), with MOSAIKS, then showed that a single set of *random* convolutional features, applied to daytime imagery, could predict a variety of outcomes (forest cover, population density, income, road length) using simple ridge regression. The finding that one feature set is predictive across many outcomes without task-specific training is the core logic that earth embeddings now generalize.

Burke et al. (2021), in *Science*, synthesized the second generation and identified the bottleneck that constrained it: every application required a specialized pipeline (image acquisition, spectral band selection, preprocessing, feature engineering, model training), and the resulting features tended to be specific to one outcome and one geographic context. The review remains the most recent comprehensive treatment of the field in a top-tier journal, and it is the natural reference point against which the embedding-based approach is evaluated. Earth embeddings address the bottleneck Burke et al. document directly, by offering general-purpose representations that do not require task-specific feature engineering.

3.3 The Applied Workflow

What does it actually look like to use an earth embedding in an applied analysis? Figure 2 shows the standard pipeline. The starting point is a set of geographic units of analysis with known coordinates. The researcher can either query embeddings directly at those coordinates, point by point, or, if the units are polygons, aggregate the embeddings over each unit's footprint. In both cases the query goes to an embedding service, typically AlphaEarth via Google Earth Engine, and returns a vector per unit. The resulting vectors are stacked with whatever conventional covariates are available, such as census variables, administrative records, nighttime lights, or prior data products, to form a single covariate matrix. That matrix is fed into a standard supervised model: ordinary least squares, random forests, gradient boosting, or a small neural network, depending on the question. The output is a predicted outcome.

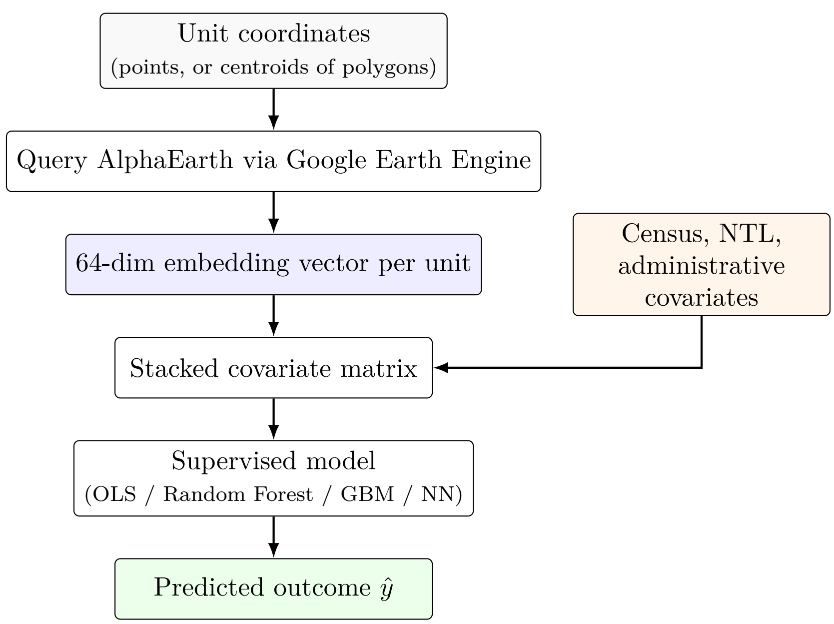

Figure 2: The applied embedding workflow. Vectors are pulled by coordinate, stacked with conventional covariates, and fed into a standard supervised model. None of the steps require deep learning or remote sensing infrastructure on the researcher's side.

The practical advantage over earlier generations is generality. The same 64-dimensional vector per unit serves as input for models predicting poverty, crime, economic activity, or infrastructure quality. There is no need to reprocess imagery or engineer task-specific features for each new application. The entry cost has dropped to a coordinate query.

3.4 Public Goods and Physical Infrastructure

Where earth embeddings have the most potential in political science today, at least on the measurement side, is the study of physically grounded outcomes, and most naturally the study of public goods provision and physical infrastructure more broadly. The claim is not that embeddings will not be useful elsewhere; it is that, given how embeddings are constructed, this is the domain where their strengths line up most cleanly with substantive questions political scientists already care about.

A brief definition. Public goods, in the strict economic sense, are goods whose consumption is both non-excludable (the provider cannot easily prevent anyone from using them) and non-rivalrous (one person's use does not meaningfully reduce availability for others). The canonical examples are things like clean air, national defense, and basic lighthouse-style services. In political science, the term is used more loosely to refer to the bundle of goods and services that states deliver to their populations where exclusion is in principle possible but is either impractical or politically unacceptable: roads, schools, hospitals, electricity grids, water systems, public safety. For present purposes what matters is that almost all of these leave physical traces on the ground. Road networks exist in space. Electrification patterns show up in imagery. The difference between a well-served and an underserved neighborhood is visible from orbit. The provision of these goods is one of the most direct observable indicators of state capacity and governance quality: a state that builds roads in some regions but not others reveals something about its priorities; a state that claims to have invested in rural electrification but whose territory shows no physical evidence of it reveals something about its credibility.

The reason this paper flirts with a broader framing, *physical infrastructure* rather than strictly *public goods*, is that many of the outcomes political scientists actually want to measure sit just beyond the formal public-goods boundary. Private housing stock, commercial building density, port activity, informal settlement growth, agricultural land use, urban form: these are not public goods, but they are physical, they vary meaningfully across space, they respond to policy, and they are visible from satellite imagery. Earth embeddings are strong at representing them for exactly the same reasons they are strong at representing public goods. Framing the target as "physically grounded features of place", with public goods provision as the most natural and politically salient example, preserves the argument without artificially narrowing it.

There is also a practical point, which Gong et al. (2026) make forcefully in their benchmark study. In many of the settings where this kind of measurement matters most, low- and middle-income countries, under-administered regions, rapidly growing informal neighborhoods, the conventional data that researchers would normally use are either missing, released on long lags, prohibitively expensive to collect at scale, or simply unreliable. In those settings, a queryable representation of what a place looks like from space is not just a methodological convenience; it is sometimes the only data source available at a comparable spatial resolution.

None of this implies that embeddings will be useless for outcomes driven more by behavior, institutions, or policy than by the physical landscape. It implies, more modestly, that the tool's near-term comparative advantage is clearer in the physically grounded space than elsewhere, and that the latter is a defensible place to start.

3.5 What We Can Say Empirically

For political scientists looking for evidence that this technology works, there are effectively two kinds of papers to read. Both come out of computer science; neither, at the time of writing, comes out of political science or economics proper.

Gong, Srivastava, et al. (2026) provide the strongest systematic benchmark so far. They evaluate three embedding families (AlphaEarth, Prithvi, Clay) for predicting 14 neighborhood-level urban indicators across six U.S. metropolitan areas from 2020 to 2023. Their central finding is that AlphaEarth's compact 64-dimensional embeddings consistently outperform the higher-dimensional alternatives. More usefully for our purposes, they make an explicit point of treating "what can embeddings predict well?" as an empirical question: predictive skill is strongest for outcomes closely tied to built-environment structure (chronic health burdens, dominant commuting modes) and weakest for outcomes shaped by fine-scale behavior or local policy (cycling rates, for instance). Performance varies markedly across cities but is stable across years, suggesting that what embeddings capture is primarily spatial structure rather than temporal dynamics. In other words, they work best exactly where public-goods-style physical features dominate the outcome, and less well for outcomes that turn on individual behavior or policy detail.

Yue, Zhao, and Hu (2026) complement this with a country-level GDP estimation pipeline that combines AlphaEarth embeddings with nighttime lights through a transfer-learning design. They train a small network to predict nighttime lights from embeddings, then use the bottleneck representation as an "income-aware" feature for downstream GDP estimation. Embedding-based estimates achieve a 16 percent lower mean squared error than nighttime lights alone for cross-sectional GDP levels, and a hybrid combination of the two does better than either, with the largest gains in countries that have weak statistical capacity. Nighttime lights remain better for short-run growth, consistent with the interpretation that embeddings capture structural features that change slowly while lights capture activity that changes faster.

Beyond these systematic benchmarks, a small number of applied papers have used earlier-generation convolutional or geospatial tools, rather than earth embeddings proper, for political-science-relevant outcomes: remote conflict monitoring (Sticher, Wegner, and Pfeifle 2023), detection and documentation of political violence from fire data and satellite images (Naghizadeh 2024), human rights documentation in zones of armed conflict (Edler et al. 2025), and public service accessibility (Huang et al. 2024). These papers are worth flagging not as examples of earth-embedding applications, but as signals of the kinds of substantive questions for which satellite-based representations have already started to be deployed. A common pattern across them is that the satellite feature enters as one block of covariates alongside conventional data, rather than as a standalone replacement. Ongoing work of my own, at the AGEB (Área Geoestadística Básica) level in Mexico City, sits inside the proper earth-embedding paradigm: AlphaEarth embeddings stacked with 2020 census variables to predict crime incidence at the neighborhood level, with the specific goal of testing whether embeddings carry information about the built environment that standard sociodemographic covariates miss.

Taken together, the empirical picture is the following: embeddings work well for outcomes with a heavy physical signature, they help but do not replace existing proxies for outcomes with a mixed signature, and they are unlikely to deliver substantial gains for outcomes driven primarily by behavior or policy rather than by the physical landscape. That pattern is consistent with the broader suggestion above, that physically grounded outcomes, with public goods provision as the most natural example, are where political scientists should begin.

4 From Prediction to Relation

We have, so far, walked through how earth embeddings are built and what they have been used to predict. Both of those are worth doing. But a political scientist reading this should, at some point, want to raise a hand and say something like: I don't actually spend most of my life predicting Y. I spend it asking how X causes Y, for specific units, under specific conditions. Can this technology do anything for *that*?

This section is the long answer to that question. Nobody, to my knowledge, has yet tried to use earth embeddings in spatial estimation, so there is no existing literature to review on the specific move this section wants to make. What can be done is to start from what the relevant literatures already know: to place the question alongside the tradition that has wrestled with it, and from there to sketch three directions in which future work might push.

4.1 Social Science Asks About Relations

The core empirical move in social science is not "predict Y from everything," it is "hold everything else fixed and ask what X does to Y." The distinction matters here because the entire machinery of Section 3, and the benchmarks in Gong et al. (2026), and the GDP pipeline in Yue et al. (2026), all live in the first world. They treat earth embeddings as rich covariates and see how far those covariates carry a supervised model toward a held-out target.

That is a reasonable thing to do, and a useful one, but it is not what political scientists usually mean when they ask about geography. When a political scientist asks whether an aid program raises schooling, or whether a minimum-wage increase affects employment, or whether a civil-war shock disrupts local markets, the quantity of interest is an *effect*: the causal impact of one variable on another for a specific unit. And once one starts asking about effects, a different question follows almost immediately, and it is the question this section is built around:

> Can the effect of X on Y for unit i be heterogeneous in the geographic character of i itself?

That is: do two units that look *structurally* different from space respond differently to the same treatment, even after we control for everything we usually think to control for? And if so, can we build that heterogeneity into our models instead of leaving it in the residual?

This is a conceptual question, not a technical one. The technical question, the one about how you actually operationalize "structurally different," is what earth embeddings let us ask in a new way. Before this technology existed, there was no good way to put "the visual, physical, morphological character of a place" into an equation, so we substituted distance, or a dummy variable, or a fixed effect. Now there is. A 64-dimensional vector summarizes the character of a place, and two places can be "similar" in a way that geography alone cannot capture.[5]

4.2 What Econometrics Did With This Question

I am not, of course, the first person to notice that geography might matter for how X affects Y. Spatial econometrics is a mature subdiscipline built almost entirely around that observation, and political methodology, following the econometrics lead closely, has spent decades importing, refining, and debating the same set of tools. But both disciplines share a feature of methodological traditions more generally: once a path has been started, the choices made early on tend to cement what comes after. The path spatial econometrics started on, for excellent historical reasons, is *distance*, and political methodology has largely followed.

The object through which this tradition encodes geographic structure is a single matrix: the *spatial weights matrix* W. Conceptually, W is an $N \times N$ table for the N units in the analysis: w_{ij} is positive when unit j is treated as a "neighbor" of unit i, and zero otherwise. The quantity $\sum_j w_{ij} z_j$, a weighted average over a unit's neighbors, is called a *spatial lag* of the variable z, and it is the basic building block of every spatial empirical model. The intuition is not exotic. A factory might pollute more when its neighbors also pollute, because regulation is lax in the area. A municipality might raise wages when neighboring municipalities raise theirs, because labor markets

[5] It is worth pausing on how strange, in a good way, this is. We have convinced ourselves that the full geographic character of a neighborhood (its buildings, streets, vegetation, land use, the texture of its built environment) can be reduced to 64 numbers. And, as the benchmarks of Section 3 show, those 64 numbers are *powerful*. It is easy to take that for granted while reading a technical description of Vision Transformers. It should not be taken for granted. This is the kind of compression that, pre-2023, would have been a research moonshot. Some of the consequences of having it, I think, we are still only beginning to see.

leak across borders. A neighborhood's crime rate might depend on its neighbors' crime, because gangs, policing, and displacement spill over. In each case the natural way to write the model is "Y_i depends on some function of the Ys or Xs of i's neighbors," and the weights w_{ij} are how the model decides who counts as a neighbor.

Two modeling families dominate, distinguished by what is being lagged. In the *spatial autoregressive* (SAR) family, the outcome is regressed on a spatial lag of *itself*:

$$y = \rho W y + X\beta + \varepsilon,$$

so y_i depends on its own covariates and on a weighted average of its neighbors' outcomes. In the *spatial-lag-of-X* (SLX) family, the outcome is regressed on a spatial lag of the *covariates*:

$$y = X\beta + WX\gamma + \varepsilon,$$

so y_i depends on its own covariates and on a weighted average of its neighbors' covariates. The *spatial Durbin model* combines the two: $y = \rho W y + X\beta + WX\gamma + \varepsilon$. Estimation is typically done by maximum likelihood (accounting for the simultaneity introduced by $\rho W y$), or by Bayesian methods (Krisztin and Piribauer 2022), or by two-stage and GMM approaches when endogeneity is a concern. The modeling toolkit is mature; what varies across applications is not the estimator but the choice of W.

And the choice of W, in the overwhelming majority of applied work, comes down to a small family of geographic specifications. *Contiguity*: units are neighbors if they share a border. *k-nearest neighbors*: each unit is connected to its k closest units by geographic distance. *Inverse distance*: the entries of W decay as $1/d_{ij}$ or $1/d_{ij}^2$ or with exponential decay $e^{-\alpha d_{ij}}$. *Distance bands* or *donut rings*: neighbors are those within a given radius, or within a band between two radii. The matrix is then almost always *row-standardized*, meaning each row is rescaled to sum to one, so that the spatial lag has a clean interpretation as a weighted average of a unit's neighbors.

Some in the field have begun to push against this default. Neumayer and Plümper (2016) argue that if spatial dependence actually runs through trade, institutional similarity, or structural resemblance, then W should reflect *that*, not just how close two centroids happen to be. Their conclusion is blunt: "models with distance or contiguity as a connectivity variable tell us little more than that the world is likely to become increasingly dissimilar the further we travel" (Neumayer and Plümper 2016, 191). Beck, Gleditsch, and Beardsley (2006) had made a similar argument a decade earlier with trade-based matrices. Cook, Hays, and Franzese (2022), in the most recent major political-methodology treatment of spatial panel data, document that even in time-series cross-section research, where *both* temporal and spatial dependence matter, the overwhelming majority of published work models only one dimension at a time: of 201 TSCS articles they review across the APSR, AJPS, and JOP between 2012 and 2019, only 12 (less than 6 percent) explicitly model both temporal and spatial dependence. They show analytically and via simulation that misspecifying either dimension biases estimates of the other and of covariate effects in general, and they propose the spatiotemporal autoregressive distributed lag (STADL) model as a general starting point for TSCS specifications. Yet even this contribution, sophisticated as it is on the estimator side and coming from as central a figure in the subfield as Franzese, leaves the specification of W itself as a conventional geographic choice: the R package that accompanies the paper automates common geographic W constructions. Wimpy, Whitten, and Williams (2021) quantify the pattern directly: 72.3 percent of spatial papers in political science use geographic distance or contiguity as the *sole* basis for W, and they argue that researchers default too readily to SAR-style models when SLX-style models, which are more flexible because they permit different Ws for different covariates, are often a better fit for the substantive theory.

A smaller but growing line of work has started to move W from a pre-declared input into something estimated, or at least disciplined, by data. Tan, Kesina, and Elhorst (2025) write $w_{ij} \propto d_{ij}^{-\alpha}$, or variants of that form, and estimate the decay exponent α jointly with the regression coefficients by maximum likelihood; different covariates can be allowed to have different decay parameters, so that infrastructure spillovers and labor-market spillovers can travel different distances. Lam and Souza (2020) estimate W as a weighted combination of candidate matrices, geographic, economic, institutional, plus a sparse adjustment via adaptive LASSO. Zhang and Yu (2018) pursue a related logic through model averaging across candidate W specifications. Qu and Lee (2015) confront the hardest version of the problem: when W itself is built from variables that respond to the outcome, standard estimators break down, and recovering spillover effects requires explicit identifying assumptions.

What this literature shares, however, is the piece most worth underlining: the candidate bases are still almost entirely geographic or economic-flow. The framework has become sophisticated. The ingredients have not. For thirty years, the empirical question of "which units influence which" has been answered from the same short menu, and the menu itself has barely changed.

4.3 Where Distance Stops Making Sense

Distance, as a way of thinking about which units influence which, makes sense in many settings and not in others. It is worth slowing down on when it does and does not, because the conceptual argument for earth embeddings rests on this distinction, and not on a blanket claim that distance is bad.

Distance makes sense when units literally affect each other through physical proximity: pollution leaks, disease spreads, labor markets bleed over, policy spillovers diffuse across borders. The reason a factory's emissions depend on its neighbors' is not that the neighbors are similar in character; it is that the neighbors are physically close. For this class of *data-generating processes* (DGPs), the mechanisms by which the data actually arise, a geographic W is not just conventional, it is correct. Every attempt to improve on it should at minimum be able to recover it as a special case.

The harder class of DGPs is the one where spatial dependence does not run through proximity at all. A few concrete examples help make the argument.

Consider first two sides of an industrial river. A residential neighborhood across the water from a factory district is separated by a hundred meters and by decades of zoning decisions. Spillovers between the two are minimal for most substantive questions. Yet a k-nearest-neighbors specification would readily identify the factory as the residential neighborhood's closest neighbor, because Euclidean distance does not know the river is there. Downtown Saint Louis and East Saint Louis, on opposite sides of the Mississippi, are the textbook version of this case: visually adjacent, substantively worlds apart, with a century of policy, industry, and race shaping the difference.

A second example is a house perched on a ridge above a city. In Euclidean terms it is very close to the valley town below, sometimes only a few hundred meters away. But the peak is wilderness while the valley is dense urban fabric. Their built environments share essentially nothing, and no serious model of how policies, shocks, or public investment propagate would treat them as the same kind of unit. A W that connects them by inverse distance is not merely inefficient; it is conceptually wrong.[6]

A third example is a gated community next to an informal settlement, two neighborhoods on opposite sides of a wall and sometimes on the same block. Their labor markets, housing markets, infrastructure, and political representation differ in ways that have almost nothing to do with how far apart their centroids sit. A spatial model that treats them as strong neighbors imports structure that is not there.[7]

In each case, the unit of analysis has characteristics, such as built form, orientation, land cover, or structural type, that a spatial model would want to use as the basis for defining neighbors, but that a classical distance-based W is blind to. This is the conceptual opening that earth embeddings walk into.

4.4 Earth Embeddings: The Missing Piece

Earth embeddings, because they compress the observable character of a place into a vector, give us, for the first time, a *data-driven representation of what a place is like.* Cosine similarity between two locations' embedding vectors is a natural, continuous measure of how much the two places resemble each other in built form, road network, land use, vegetation, and texture, regardless of how far apart they are on a map. An embedding-derived W, one where w_{ij} rises with embedding

[6] Mexico City provides a clean version of this example. The valley of Mexico is one of the densest urban fabrics in the world, surrounded by mountains, some of them inhabited at the margins by communities whose economic, political, and administrative life has little in common with the city they overlook. A k-nearest-neighbors matrix built on Euclidean distance would cheerfully mix them together.

[7] Favela Rocinha in Rio de Janeiro is perhaps the best-known example of this pattern: a dense informal settlement bordered by affluent, formally planned neighborhoods such as São Conrado and Gávea. The Euclidean distance between the two is sometimes measured in meters; the substantive distance in terms of state presence, housing markets, and public-service provision is enormous.

similarity, is exactly the kind of "non-geographic candidate basis" that the literature on estimating W from data has been asking for without quite saying.

To my knowledge, no peer-reviewed paper has yet proposed using earth embeddings to build W. This is not a claim of originality. Earth embeddings, as a cleanly defined and usable object of research, are essentially brand new; the ecosystem has only recently consolidated into something an applied researcher can actually reach for. The gap is open because the raw material only recently arrived, not because anyone was looking past it. What the moment offers, then, is an unusually clear opportunity: the technology is mature enough to use, and the methodological question it could help answer has been waiting for something like it for decades.

4.5 Three Directions for Future Work

If a future researcher were to take the step of bringing earth embeddings into spatial estimation, three directions seem to me most natural. None are solved; they are research programs.

Direction 1: A hybrid weights matrix $W = f(\mathit{distance}, \mathit{similarity})$. Rather than pick between geography and embedding similarity, one could build a new kind of W that combines them through a learned function, trained jointly with the regression coefficients. In the simplest version, w_{ij} would depend on some flexible combination $f(d_{ij}, \mathrm{sim}(\mathbf{e}_i, \mathbf{e}_j))$, with the functional form and any parameters of f estimated from the data. In a more ambitious version, one would go back to the embedding architecture itself and train a "vitamin-enhanced" location encoder that outputs a vector jointly informed by geographic position and structural similarity, so that the resulting W is the native output of a model designed for spatial estimation rather than a bolt-on.

Direction 2: An additional similarity matrix S alongside W. Rather than replace the distance-based W, future work could add a second matrix, call it S, built from embedding similarity, and include both in the model. In a Durbin-style specification this would look like $y = \rho W y + \lambda S y + X\beta + WX\gamma + SX\delta + \varepsilon$, so that distance-driven spillovers and similarity-driven spillovers each carry their own coefficient. The estimator would then report, rather than assume, how much of the spatial structure is doing geographic work and how much is doing similarity work. This is the most incremental of the three directions and, probably for that reason, the most likely place to start.

Direction 3: Similarity-based regularization of W. A third option is to keep W defined geographically but to penalize or reweight its entries on the basis of embedding similarity. Two units that are close in distance but far in embedding space (the mountain peak and the valley town; the gated community and the informal settlement) would have their w_{ij} reduced, effectively telling the model "they're physically close, but they are not the same kind of place, and the spillover should be downweighted." This is a way of smuggling similarity into a geographic W without writing a new estimator, and it fits naturally into the penalized-W framework Lam and Souza (2020) already use.

The research agenda, then, has two arms. The first is conceptual: *which* data-generating processes is a similarity-based W (or S) actually appropriate for? Some questions (local labor markets, pollution spillovers) are fundamentally about proximity, and adding similarity will not help them. Others (policy diffusion across structurally comparable jurisdictions, epidemiological DGPs that depend on landscape type) are fundamentally about similarity, and distance is the wrong primitive. A real theory of when each matters is missing. The second is methodological: *how* should similarity actually enter the estimation, with what identifying assumptions, and with what practical recipe a researcher can follow? Both arms are open. The contribution of this paper is to argue that the question is worth asking and that the raw material for asking it now exists.

Before closing this section, one objection is worth pre-empting, because anyone who has followed the argument this far will have thought of it: does this mean throwing out geography? It does not. Similarity is a second ingredient, not a replacement, and the reasons geography still matters are not subtle.[8]

5 Challenges and Limitations

Earth embeddings are a useful tool for measurement and a plausible input into spatial estimation. Before turning to the research agenda, it is worth being explicit about what they do not do well, and where the limitations of the current technology constrain what can be built on top of it.

5.1 Construct Validity

Earth embeddings capture the visual and environmental characteristics of a location as observed by satellite sensors. Whether that is informative about political and institutional outcomes depends on the extent to which those outcomes leave physical traces. For outcomes like infrastructure quality, economic activity, and urbanization, the connection is strong. For outcomes like institutional trust, elite bargaining, or political preferences, the connection is less direct. That said, a narrowly visual reading of embeddings may underestimate their reach. Physical infrastructure, the condition of public buildings, road quality, and the presence or absence of services are not merely correlates of governance; they are, in many cases, revealed consequences of state behavior.

[8]Consider an embedding-only W built on cosine similarity between AlphaEarth vectors. Two dense European capitals, say downtown Paris and downtown Madrid, will look extraordinarily similar in embedding space: comparable density, comparable street morphology, comparable mix of built form and green space. An embedding-similarity W would connect them with a high weight, treating them as close neighbors for the purposes of a spatial regression. For most spatial questions this would be wrong, in ways that are not subtle: they are in different countries, under different legal and fiscal regimes, with different labor markets, and spillovers across that gap are nothing like spillovers between two adjacent neighborhoods in either city. The point of this section is not that geographic W should be discarded. It is that similarity is a second, genuinely new ingredient that classical W has never had access to, and that future work should be about *combining* the two rather than picking between them. Directions 1 through 3 above are three different ways of doing that combining.

5.2 Interpretability

Unlike hand-crafted features such as NDVI (vegetation index) or nighttime light intensity, where each value has a direct physical interpretation, embedding dimensions have no individual meaning. When an embedding-based model predicts high poverty for a location, the researcher cannot straightforwardly identify which visual features drove the prediction. There is an active line of machine-learning research aimed at addressing this, using tools such as probing classifiers (which train simple models to predict interpretable quantities from embeddings), attention visualization (which highlights the image regions most responsible for a given embedding), and documentation standards such as the Embedding Cards proposal in Klemmer et al. (2025). That work, however, is properly the subject of a different paper; for the purposes of this review, interpretability is best read as an open limitation that political scientists using earth embeddings will inherit, not solve, from the foundation-model literature.

5.3 Spatial Bias in Training Data

Earth embedding models are not trained on uniformly distributed data. More and higher-quality imagery is available for Europe, North America, and parts of East Asia than for Sub-Saharan Africa, Central Asia, or small island states. Urban areas are overrepresented relative to rural ones. Embeddings may therefore be more informative for well-represented regions and less reliable elsewhere (Purohit et al. 2025). This matters especially for comparative research across diverse contexts, which is precisely the research setting where embeddings would be most valuable.

5.4 Scope: A Data Layer, Not a Research Design

Earth embeddings are a data layer. They provide richer features and potentially more informed spatial relationships for political science research, but they are not a substitute for sound research design. A well-specified embedding does not rescue a poorly identified model, just as a well-coded variable does not rescue a confounded regression. Everything in this paper is conditional on the quality of the empirical work that builds on it. The point of bringing embeddings into political science is not to do empirical work *differently*; it is to do it with a richer notion of what geography is.

6 Conclusion

Three arguments, one technology. This paper has reviewed what earth embeddings are, what they have been used to measure, and one direction they open for social science research. Putting the three together, the picture is this. There now exists a cheap, queryable, general-purpose representation of what any geographic location looks like from space. The representation was built by computer scientists, and the benchmarks that evaluate it have been built inside computer science. Empirical social science has not yet picked it up at scale, which is unsurprising given how new it is, but this gap is a genuine research opportunity rather than a permanent state of affairs. The near-term home with the most potential in political science is the measurement of physically grounded outcomes, with public goods provision and physical infrastructure as the most natural examples, because those are the kinds of features embeddings are built to see. The farther-term opening is on the estimation side, specifically in the long-running effort to specify spatial weights matrices that reflect how places actually relate to each other rather than just how far apart they sit on a map.

A larger point is worth keeping in view. The fact that the full geographic character of a place, its roads, buildings, vegetation, land use, and spatial organization, can be reduced to a vector of a few dozen numbers is, in the strict sense, revolutionary. Those numbers are already enough, as the benchmarks show, to predict much of what has historically been measured from space. They are potentially enough, as this paper has argued, to reshape how geographic interdependence is modeled in causal estimation. And, as with any technological change this new, how far it can go is not yet known. What this paper has tried to do is lay out the current state, point at the openings, and argue that political science has both something to gain from walking through them and something to contribute if it does.

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